MID-

The mid-range category consists of nine motherboards for the Intel platform that are priced between Rs 5,500 and Rs 10,000. Only one motherboard—the ASUS P4S800-MX—was based on the SiS chipset; the other eight are based on Intel's 865 and 845GE chipsets. The 865 chipset is new, and outperforms the ageing 845GE-based motherboards. It was quite clear that the competition would focus between the 865-based chipsets and the lone ASUS SiS 661FX chipset. Both support the latest features, so it's only performance that sets them apart.

Features

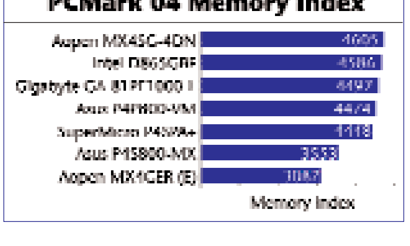
The most important aspect of a motherboard is the features set; this decides how long your motherboard will be of service to you. If you buy a board with the latest features, you have a better chance of still being able to use it two or three years down the line. In the ever-evolving IT industry, things change fast, and you certainly don't want to buy a motherboard that could be obsolete in a year.

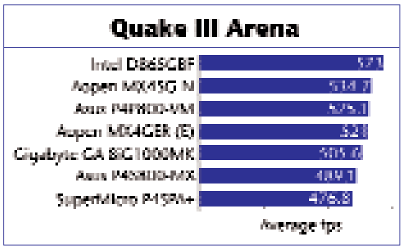
The Intel 865-based motherboards offer the best headroom in terms of features. The upgrade options are numerous, with support for 800 MHz FSB, dual channel DDR 400 MHz memory modules, support for eight USB ports and the onboard native SATA controller, along with the IDE connectors, means you have upgrade options in all the three critical sub-systems-processor, RAM and hard drives. Since the 800 MHz FSB processors are still expensive, you can opt for a 533 MHz FSB processor with Hyperthreading, and upgrade when prices fall. You can do the same for RAM modules and hard disks. The SiS 661FX has all the same features, except for dual-channel memory mode.

The 865PE chipset-based Gigabyte GA-81PE1000-L was the only motherboard in the mid-range category that did not have onboard graphics

Quite a few boards offered exceptional features. The SuperMicro P4SPA+ and Gigabyte's GA-81PE1000-L offered just about everything you would ever want. Both are built around the Full ATX form factor, and the components are well spaced out, except the oddly-placed power connector in the SuperMicro-the connector is hard to remove once fixed. A close competitor to these two was the Intel 865GBF, but it lost on features due to the omission of an Ethernet controller—the SuperMicro has gigabit Ethernet.

In the Micro ATX form factor segment, the MX4SG-4DN was the better of the two boards from AOpen—the 845 chipset-based AOpen MX4GER only comes with two memory slots. The ASUS P4P800 and Gigabyte GA 8iG1000MK packed everything into a small package and at decent prices. All these boards support 800 MHz FSB, 400 MHz memory frequencies and SATA ports; none came with more than two SATA ports, but for their prices, this isn't disappointing.





The Gigabyte boards were very colourful, and for a purpose—the colour coding makes installation as fool-proof as possible. Most of the 865 boards, except the Intel 865GBF and SuperMicro boards, offer BIOS tweaking for overclocking. Overall, the ASUS P4P800 and the AOpen MX4SG-4DN came out tops, and offer the best features.

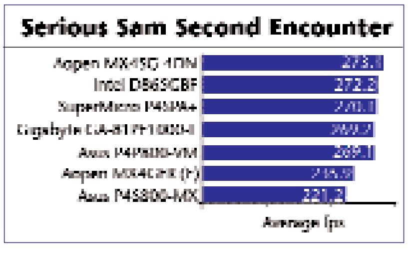
Performance

In the gaming benchmarks, the Intel 865GBF motherboard beat the lot with its great performance. It managed to stay ahead of competition in all three gaming tests. The AOpen MX4SG-N and ASUS P4P800 followed, and were good at their job. The ASUS P4S800 motherboard lacked the punch required, and came in last, due to the low-memory bandwidth it has when compared to the 865-based boards.

In the real-world application benchmarks (Ziff Davis Business Winstone, Multimedia Content creation and video encoding), the AOpen MX4SG-4DN edged ahead of its closest competitor, the Gigabyte GA-81PE1000-L, by just 0.32 units. ASUS came in a close third, just 0.42 points behind the AOpen board. The AOpen MX4GER came in last in this test, mainly due to its 845 chipset that was outdone by the rest.

In the synthetic benchmarks, the Intel 865GBF raced ahead again. Coming in a very close second, was the Gigabyte GA-81PE1000-L. The Gigabyte board was just five points behind the Intel board's score of 3,395 in the 3D Mark 2003 test. The AOpen MX4GER sank to the bottom, proving conclusively that an overpriced 845 chipset-based motherboard has no business even competing in this category.

The 865-based boards blew the competition away. The ASUS P4S800, based on the SiS661FX chipset, performed well against the 865, but lost due to its single-channel memory sub-system. The higher bandwidth that the 865s enjoy, due to their dual-channel memory sub-system, spelt disaster for the SiS-based board. What's more, boards based on both chipsets are priced alike, making it unthinkable of buying the SiS661FX-based board. Another unbelievable blunder is the pricing of the 845-based AOpen MX4GER; who would pay Rs 5,000 for a board that performs poorly and has no upgradeability?



Overall, in terms